RAJALAKSHMI ENGINEERING COLLEGE

Prakashini S 251801231-2025-AIDS

P2

| | |
|---|---|
| **Started on** | Thursday, 24 September 2026, 10:26 AM |
| **State** | Finished |
| **Completed on** | Thursday, 24 September 2026, 10:31 AM |
| **Time taken** | 5 mins 32 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:** (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0; i<n; i++){
        scanf("%d",&arr[i]);
    }
    for(int i=0; i<n;i++){
        for(int j=i+1; j<n ; j++){
            if(arr[i]==arr[j]){
                printf("%d",arr[i]);
            }
        }
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | ✔ |
| ✔ | 5<br>1 2 3 4 4 | 4 | 4 | ✔ |
| ✔ | 5<br>1 1 2 3 4 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

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P2

| | |
|---|---|
| **Started on** | Thursday, 24 September 2026, 10:32 AM |
| **State** | Finished |
| **Completed on** | Thursday, 24 September 2026, 10:43 AM |
| **Time taken** | 10 mins 45 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];
    int count[n + 1];

    for (int i = 0; i <= n; i++) {
        count[i] = 0;
    }

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        count[arr[i]]++;
    }

    for (int i = 1; i <= n; i++) {
        if (count[i] > 1) {
            printf("%d\n", i);
            break;
        }
    }

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | ✔ |
| ✔ | 5<br>1 2 3 4 4 | 4 | 4 | ✔ |
| ✔ | 5<br>1 1 2 3 4 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Prakashini S 251801231-2025-AIDS

P2

| | |
|---|---|
| **Started on** | Thursday, 24 September 2026, 10:47 AM |
| **State** | Finished |
| **Completed on** | Thursday, 24 September 2026, 11:23 AM |
| **Time taken** | 35 mins 52 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
|---|---|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int t;
    scanf("%d", &t);

    while (t--) {
        int n1;
        if (scanf("%d", &n1) != 1) break;
        int a[n1];
        for (int i = 0; i < n1; i++) {
            scanf("%d", &a[i]);
        }

        int n2;
        if (scanf("%d", &n2) != 1) break;
        int b[n2];
        for (int i = 0; i < n2; i++) {
            scanf("%d", &b[i]);
        }

        int first = 1;
```

```
        int first = 1;
        for (int i = 0; i < n1; i++) {
            for (int j = 0; j < n2; j++) {
                if (a[i] == b[j]) {
                    if (!first) printf(" ");
                    printf("%d", a[i]);
                    first = 0;
                    break;
                }
            }
        }
        printf("\n");
    }

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |
| ✔ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Prakashini S 251801231-2025-AIDS

P2

| | |
|---|---|
| **Started on** | Thursday, 24 September 2026, 11:23 AM |
| **State** | Finished |
| **Completed on** | Thursday, 24 September 2026, 11:24 AM |
| **Time taken** | 52 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
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Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
|---|---|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:** (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int t;
    scanf("%d", &t);

    while (t--) {
        int n1;
        if (scanf("%d", &n1) != 1) break;
        int a[n1];
        for (int i = 0; i < n1; i++) {
            scanf("%d", &a[i]);
        }

        int n2;
        if (scanf("%d", &n2) != 1) break;
        int b[n2];
        for (int i = 0; i < n2; i++) {
            scanf("%d", &b[i]);
        }

        int i = 0, j = 0, first = 1;
```

```
        while (i < n1 && j < n2) {
            if (a[i] < b[j]) {
                i++;
            } else if (b[j] < a[i]) {
                j++;
            } else {
                if (!first) printf(" ");
                printf("%d", a[i]);
                first = 0;
                i++;
                j++;
            }
        }
        printf("\n");
    }

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |
| ✔ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Prakashini S 251801231-2025-AIDS

P2

| | |
|---|---|
| **Started on** | Thursday, 24 September 2026, 11:26 AM |
| **State** | Finished |
| **Completed on** | Thursday, 24 September 2026, 11:32 AM |
| **Time taken** | 5 mins 30 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

**For example:**

| Input | Result |
|---|---|
| 3<br>1 3 5<br>4 | 1 |

**Answer:** (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    int k;
    scanf("%d", &k);

    int found = 0;
    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (a[j] - a[i] == k || a[i] - a[j] == k) {
                found = 1;
                break;
            }
        }
        if (found) break;
    }

    printf("%d\n", found);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 3 5<br>4 | 1 | 1 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1 | 1 | 1 | ✔ |
| ✔ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0 | 0 | 0 | ✔ |
| ✔ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Prakashini S 251801231-2025-AIDS

P2

| | |
|---|---|
| **Started on** | Thursday, 24 September 2026, 11:32 AM |
| **State** | Finished |
| **Completed on** | Thursday, 24 September 2026, 11:36 AM |
| **Time taken** | 3 mins 34 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

**For example:**

| Input | Result |
|---|---|
| 3<br>1 3 5<br>4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    int k;
    scanf("%d", &k);

    int i = 0, j = 1;
    int found = 0;

    while (i < n && j < n) {
        if (i != j && a[j] - a[i] == k) {
            found = 1;
            break;
        } else if (a[j] - a[i] < k) {
            j++;
        } else {
            i++;
        }
        if (i == j) {
            j++;
        }
    }

    printf("%d\n", found);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 3 5<br>4 | 1 | 1 | ✔ |
| ✔ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1 | 1 | 1 | ✔ |
| ✔ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0 | 0 | 0 | ✔ |
| ✔ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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